

Research Assignment 1

Section A : Data Fundamentals

1. A database is a system where data / information is stored. It's basically like a well-organised filing cabinet made to store, manage and retrieve large amounts of information.

- Databases are essential in the modern world because:

- * They can store large amounts of information effectively
- * They are quick and can retrieve a single piece of data in a large spreadsheet quickly
- * They allow for easy analysis of information and reporting where reports can be created and decisions can be made

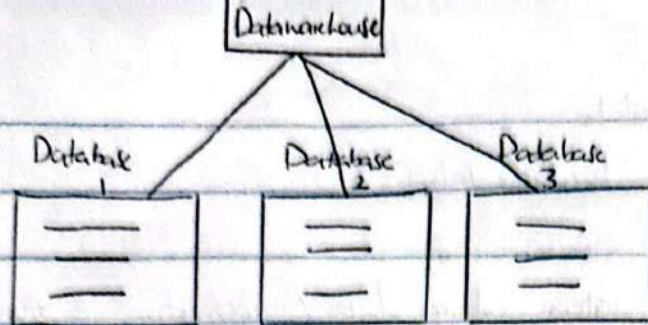
Examples of where databases are used:

- Fitness Trackers - When you exercise, data such as steps, calories burnt is stored in a database
- In Social media when you ~~enter~~ like, post or follow someone, that information is stored in a database
- In Online transactions (where you can pay and they already have all your information e.g bank details, access money in your account, update bank balance after making a transaction)

2. A datawarehouse is a system that stores multiple databases from different sources made for reporting and analysis. It stores clean & structured data.

- How does it differ from a regular database?

- * It analyses historical data / transactions
- * A regular database records daily operations / transactions using OLTP and a data warehouse stores large amounts of data / information from multiple sources and uses OLAP to analyse that information and ~~interpret it~~ ^{report}



Why would a company use a datawarehouse?

- * They would use it to make huge smart decisions, predict trends after ^{analysing} using historical, reliable data
- * To Keep large amounts of information so it doesn't get lost

3. A datalake is used for storing all sorts of data (Structured / semi-structured and structured data)

- It stores raw, unprocessed data until its needed
- Whereas a datawarehouse stores structured data that is clean

A DataLake is like a lake that has large amounts of water that is not clean

- A Data warehouse is like in a bottled water factory where water comes on and gets filtered, cleaned and packaged then sold to consumers for drinking
- A business would choose a data lake over a data warehouse when their operational needs require advanced analytics, flexibility.
- Another difference between the 2 are the people who use them (Target market)
Since datalakes are unorganized and complex, they are mostly used by data scientists, and data engineers who have skills required to analyse those messy files
- Data warehouses are clean and organized so their users are data analysts, Business Analysts and Executives who need answers

4. A data mart is a small storage of data made for a specific department / team in a company, it provides focused, relevant data for those specific departments quicker

Data marts basically save time, they take a lot of data and make it easy to use, Teams can make decisions based on the data without

needing assistance

- A data warehouse is the parent, It's like a library for the whole organization, The datawarehouse gets data from every department like HR, Sales, Finance, it pulls all this data in one system
- The data mart is the child, Since the data warehouse is really huge and hard to use, smaller parts of the data are taken out to make data marts, A data mart is like a book-shelf that has only the data that one team needs.
- It is usually used by Managers / Team Leaders, Business and Data analysts and Executives
- * The Data warehouse serves the entire company, Data Marts serve specific departments like Sales / marketing team / HR

5. Difference between Structured, semi-structured data and Unstructured data and give an example of each?

Structured Data	Semi-structured data	Unstructured data
* Organized and neat	* Not as neat as structured	* Raw data that has not
* Can be stored in tables, rows and columns for analysis	data and cannot be stored into columns, rows but not as chaotic as unstructured data	been processed in any form and doesnot fit into tables or columns
* Easy to search and work with	* Needs more work to be processed	
* Example - Records for transactions in a spreadsheet / sales, Database tables	Examples - Emails, social media posts	Examples - Images, Videos, Audio files

6. Data Mining - Process of uncovering hidden patterns, trends, correlations and Insights from a large datasets that are not easily visible through normal observation

USES IN REAL WORLD include:

- * Reducing fraud - By detecting unusual transaction patterns
- * Increase sales - Identify which customers are likely to buy next
- * Netflix recommendations - Data mining is used here to show/suggest movies/shows based on what you watched, how much you watched, search behaviour and preferences. These insights are then used to improve the whole experience for the user

7. Data Modeling

- A process of designing how data will be stored, organised and related to each other before building anything
- * It prevents data quality problems early / eliminates errors
- * It improves system performance
- * Act as a translator - By making sure that the business and the IT people are on the same page

TYPES of Data Models

1. Conceptual Data Model

- Designed for non-technical stakeholders, it is the highest ^{level of} modeling
- It shows what data the system contains and how business concepts relate
- It does not involve technical details and database rules

2. Physical Data Model

- This is the most detailed model which is tailored to a specific database system
- It includes primary keys, foreign keys, indexes and constraints
- Used by database administrators to build the actual database

Examples: Code scripts for creating tables and setting storage optimizations

8) A Database Management System (DBMS) is a software to interact with databases and access the data without the DBMS you would just have raw data sitting on a drive / memory card with no way to access it

An example is: The Book management System (Digital Library Catalogue)

- * At a university or local library, you use a computer terminal to find books. The database lists all books with details like author, genre and location, The librarian uses the DBMS to manage this
- * So when you search "Science" books, the DBMS shows only those books
- * When a book is checked out, its status updates to "Borrowed" to prevent others from accessing it

FOUR POPULAR DBMS platforms

- MySQL - One of the most popular platforms which is widely used and free and open to everyone in the world, it stores data in neat rows and tables
 - Suitable Use - For Websites and Web applications that require a reliable, fast and simple database
- E.g. A small business (online) can use MySQL to store product listings, customer orders and details

- Oracle Database - A Powerful database built for very large organisations. It's not free but comes with world-class performance and security.

- Suitable Use - Government Institutions and Large Corporations that handle large amounts of critical data

E.g. Huge banks like FNB, Capitec uses Oracle to manage millions of customers

- Mongo DB - It stores data in flexible documents not in tables and rows, This makes it great for unstructured or constantly changing data

SUITABLE USE - Modern apps where data structure changes all the time / not predictable

E.g. A social media app can use MongoDB to store user profiles, posts, comments, likes

• PostgreSQL - An advanced open source database, similar to MySQL but with more powerful and advanced features and better handling of complex data

SUITABLE USE - Applications that require advanced queries, large data or more complex logic

E.g. A huge company like Google uses PostgreSQL to process large amounts of data.

9 ETL is a process by which the data moves from the database to the warehouse

* Extract - Raw data is pulled from various databases (e.g. websites, spreadsheets, social media, sales system)

* Transform - Data is cleaned, standardized and shaped to make it consistent and useful (e.g. Removing duplicates, fix errors and missing values) and removal of unnecessary information

* LOAD - The Clean data is moved into the data warehouse once it has been loaded, it is ready for analysis, reporting and decision making

WHY is this process important in Data Analytics

* It saves time - It automates data cleaning, allowing data analysts to focus on analysing and making decisions

* It Unifies the data from different departments, ensuring consistent numbers for everyone

10. Common Data Storage Formats in Data Analytics are:

* CSV (Comma-Separated Values)

- Simplest, most commonly used data ^{format} storing data as plain text, with each row as a record and values separated by commas

Example: Name, Age, City

Sanele, 23, Johannesburg

Amahle, 21, Durban

- Best for quick data export, sharing small datasets with non-technical users

* JSON (JavaScript Object Notation)

- It is a text-based, semi-structured format using key value pairs and nested arrays

- It is Best for web apps or complex data structures that may change over time

* AVRO - A row based storage format with an embedded schema that describes the data alongside the data itself

* Best to use when there is real-time data streaming

- Perfect for real time data pipelines and frequent full record writes

* PARQUET - Column oriented, binary format for big data analytics

- It stores data column by column which speeds up queries by reading only needed columns

- Should be used when handling large datasets in data warehouses/lakes

CSV - Used for simplicity and for small-medium datasets

JSON - For complex, nested data structures

AVRO - For real time data movement and streaming use cases requiring self describing formats

Parquet - For large datasets and serious analytics which require speed and storage

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SECTION B

- 11 AI - The ability of computers/machines to think, learn and perform tasks that usually require human intelligence

Narrow AI (Weak AI)

- * Type of AI trained to perform one specific task very well, but unable to do anything beyond that task

Example: Google Translate which can only translate languages but cannot perform other unrelated tasks

General AI (Strong AI)

- * AI that can think, reason, learn and perform any task and can switch tasks

~~Example~~ Strong AI - does not exist yet, we're still using Narrow AI

Difference: Strong AI is versatile and can perform any intellectual task whereas Weak AI is bound to a specific task

- * Weak AI relies on large amounts of specific data to master a task. Strong AI can learn from just a few experiences and transfer that knowledge to completely different areas

- 12) Machine Learning - A field of AI where machines learn from data on their own, without step by step instructions

- How does it differ from traditional programming - In traditional programming, programmers explicitly write fixed rules that tell the computer exactly how to process data to produce a result whereas in machine learning you provide the computer with data and the expected outcomes and the algorithm automatically discovers the rules by itself

Supervised Learning - The machine learns from labelled data where the correct answer is known, it compares its predictions to the correct answers and improves over time

Unsupervised Learning - The machine learns from unlabeled data without guidance, it explores data to find hidden patterns

Reinforcement Learning - Machine learns by trial & error, while receiving rewards for good decisions and penalties for bad ones, it improves by interacting with its environment rather than using a fixed dataset

13 Deep Learning - a specialized area of Machine Learning where machines learn from data using neural networks, structures inspired by the human brain. This approach allows Deep Learning to excel at complex tasks like face recognition and speech understanding which basic machine learning often struggles with

How Deep learning relates - Artificial Intelligence is the broad goal of creating machines that think and act intelligently

- Machine Learning is a field of AI that allows machines to learn from data instead of following explicit programming. Within machine learning, Deep learning is a specialized powerful technique using multi-layered neural networks to learn from extremely large and complex data

- Neural Networks - An engine that powers Deep learning

It's a system of interconnected layers of mathematical nodes called neurons (they mimic how the human brain processes information)

- Neural Networks are made up of 3 different layers

* The Input Layer (Layer 1) - Where raw data enters the network
This layer does not do any thinking, it just receives the information and passes it forward

* Hidden layers (Layer 2) - This is where deep learning happens

- Hidden layers receive input from either the input or previous hidden layers, A neural network can have many hidden layers

- Each layer further processes the data received from the previous layer before passing it along

* Output Layer - (Layer 3)

The output layer produces the final result of the neural network's processing

- * When a neural network is created, it starts with no knowledge, there's just connections assigned random weights
- * During training, the network processes labeled examples, makes predictions, compares them to the correct answers and measures error
- * It then uses backpropagation to adjust the weights backward through the layers, reducing errors step by step. This process keeps on repeating over many examples, eventually fine tuning the weights so the network can accurately predict outcomes on new data

14 Natural Language Processing (NLP) - A field of AI that helps computers understand, interpret and generate human language, both written and spoken

Examples include:

- * Search Engines like Google
- * Email spam filters which classify emails as spam or legit by analyzing words and context and the sender's behaviour
- * Virtual Assistants like Siri

15 LLM - an AI model that has been trained on huge amounts of text data that can perform diverse language tasks such as writing, reading, translating and answering questions in human language

- * The difference between the 2 is that LLM has been trained using ^{lots of} unlabelled text from the internet, code or books whereas with the standard machine learning model, it has been trained with structured, labelled datasets
- An LLM can adapt to many tasks with just short instructions

- A standard ML model is good at one thing, but useless outside it
- Examples of 2 well-known LLMs - Claude which is used for research assistance, explaining difficult concepts, drafting professional content and analysing data
- * Another example is ChatGPT which is similar to Claude and also used for writing assistance and explaining difficult concepts

16 Generative AI

- AI that can generate new content such as text, images or other media
- Generative AI works at high level by firstly learning the patterns, it is trained on lots of data (images, text, audio) it learns underlying structures and relationships without memorizing the exact data
- It then receives a prompt from a User (it could be a sentence, image, question / instruction)
- It then generates / create relevant content that can be a text, image using the knowledge it learned.

3 Models 3 Tools that Use Generative AI

- * ChatGPT - Generates human like responses, computer code and other insights
- Used to brainstorm, respond to emails, summarize long reports and explain difficult concepts
- * Midjourney - It creates images from text descriptions, mostly used by designers and artists for creative projects
- * GitHub Copilot - AI assistant for developers that suggests code, which speeds up development by automating repetitive tasks

17 Training Data Set - A collection of data used to teach AI how to perform tasks

- * The quality of the data matters because:
- Errors in data / missing values can cause AI to learn incorrect information, leading to inaccurate results

* AI reflects the data it trains on, skewed datasets cause biased models such as facial recognition systems performing poorly on underrepresented groups

A lot can go wrong if the training data is incorrect

* Incorrect data in critical systems like medical diagnosis can cause harm to people's lives

* Biased data leads to biased outcomes/output, for example a facial recognition system trained mostly on light-skinned faces can perform poorly on darker skinned faces leading to unfair consequences

18) Computer Vision - A field of AI focused on teaching computers to interpret and understand visual information

- Computers perceive images as grids of numbers called pixels

Grey-scale images are represented by a matrix where each pixel has a brightness value from 0 (black) to 255 (white)

- Color images have 3 layers, Red, green and Blue, each represented by separate matrices showing color intensity at each pixel

- The system then analyzes images in layers using CNNs (Convolutional Neural Networks), ^{Early} layers identify simple features like edges and contrasts

- Middle layers combine these into shapes and textures

- Deep layers recognize complex features such as (eyes, faces, words, wheels) to accurately classify the image content

- The interpreted results trigger actions such as labeling, alerts, phone unlocking, etc.

A REAL WORLD USE EXAMPLE

Barcode / QR Code Scanning

* Cashiers scanning groceries rely on computer vision to interpret visual patterns and convert them into meaningful data instantly

Face Unlock on Smartphones

* The phone camera captures your face, maps key facial landmarks

(nose, eyes) and matches it to your stored profile in a few seconds to unlock the phone

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- Section C: APIs, Tools & Developer Concepts

19) API - Set of rules which allow different applications to communicate. It acts as a digital bridge, setting clear rules and formats so one system can request information from another without knowing its internal workings.

A real-life Example of APIs:

The Airport Check-In-Desk is a good example of how an API works

- So you are the app/user making a request, The Airline's Internal system holds/store all booking, seat, passport and baggage information
- The check in desk agent acts the API
- You don't get direct access to the airline's database, but you hand over your passport and booking reference to the agent who queries the system, verifies the information and gives you the boarding pass with all the details needed

Why are APIs important in data systems?

- They save a lot of time for developers (who use existing APIs like Stripe for payments or weather APIs instead of building features from scratch)

- They keep Systems secure - They serve as controlled gateways, exposing only selected data with authentication and access controls and protect internal systems
- They allow systems to work together - By connecting many specialized tools via APIs e.g. a Bolt app (e-hailing) integrates maps, payments and SMS APIs all together to provide the best experience for the user

20 API Key - Secret, Unique code consisting of a string of letters and numbers that identifies and authenticates a person or project when making requests to an API

Purpose of an API Key - Tracks how much the user consumes, enforces limits and calculates billing by logging every call made with the key

- * Identification purposes - Identify who is making the request
- * Authentication and Access control

How it should be stored and protected?

- * Use secret management tools that encrypt keys and securely provide them to your application at runtime (For platforms like AWS, Azure)
- * Protect Version control - Add configuration files containing keys to ".gitignore" text file to prevent accidentally uploading to public repositories
- * Never share keys over Insecure channels like email
- * If an API Key is exposed publicly - attackers can rack up large bills by making thousands of requests using your API Key linked to paid services
- * Attackers may access sensitive data without detection as requests appear legit
- * Data deletion / Corruption
- * Identity Abuse / Impersonation

21) REST API - A type of API that follows specific architectural rules for communication over the Internet, mostly used in web & mobile development

- * REST stands for Representational State Transfer

Main HTTP Methods in REST APIs

* GET (Read/Retrieve data)

It requests data from the server without changing it (Used for fetching information like social media feeds)

* POST (create new media)

Sends new data to the server to create a new resource like registering an account

* PUT (Update Existing Data) - Updates or replaces an existing resource

* DELETE (Remove data) - Deletes a resource permanently such as deleting an account/removing a photo

22) JSON - A lightweight format for storing and transporting data

- It is commonly used when working with APIs because:

- * It is Universally supported - Every major programming language offers built in libraries to generate JSON making it perfect for seamless data exchange between different languages

- * It is lightweight - It uses minimal syntax

- * It is clean and easy to read, allowing developers to understand data

Simple Example of a JSON object

```
{ "item_name" : "Apples",  
  "quantity" : 6,  
  "Price_per_kg" : 24.50,  
  "is_organic" : true,  
  "Store_aisle" : "Fresh Produce" }
```

A list of items in a grocery shopping list

- * These {} mark the beginning and end of our data container

- * Item name is the Key and a value is on the right separated by a colon "Apples"

23) Prompt Engineering - A way to construct an effective prompt to ensure clarity to the AI so that it doesn't give you the wrong answer

→ Prompt Engineering is an Important skill to have because:

- * It saves you time - A well crafted prompt reduces back and forth clarifications, you get useful answers faster
- * It becomes a skill in the workplace if you integrates it into industries like marketing, healthcare, finance and law effectively

* Examples of a poorly-written Prompt and an improved version

Poor Prompt - "Explain what coding is"

Improved prompt - "Explain what coding is to a complete beginner who has never written code before. Use a simple-world analogy to introduce the concept then show a basic example in Python. Keep the tone friendly and avoid using technical jargon"

Poor Prompt - "Write something about social media/caption"

Improved prompt - "Write a 300-word Instagram caption strategy guide for a small South African clothing brand targeted to women aged 18-30. Include hashtags and how to drive engagement."

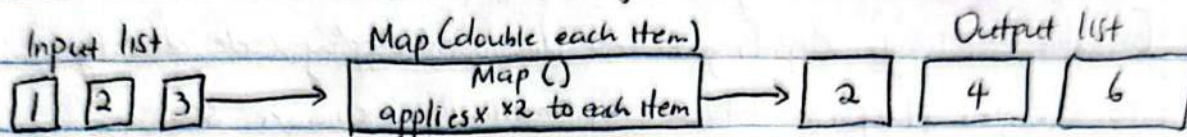
24) MAP Function - a programming feature that applies a given operation or transformation to every item in a collection and returns a new collection with the modified results

- * It streamlines repetitive tasks by automatically processing each element without requiring explicit loops

- In programming it accepts a function and a list (arrays) as inputs
- Executes the function on each item in the list

- Generates a new list with the transformed items
- Simplifies code by avoiding manual loops, applying the same operation to every element automatically

Example of a diagram



Simple Python Example

A list of numbers

```
numbers = [1, 2, 3, 4, 5]
```

Define what transformation to apply

```
def double(x):
```

```
    return x * 2
```

Apply it to every item using map()

```
result = list(map(double, numbers))
```

```
print(result)
```

Output : [2, 4, 6, 8, 10]

25 Cloud Computing - The ondemand provision of computer resources (such as infrastructure & storage) as on-line services, it removes the requirement that people and companies manage their own physical resources and only pay for what they really use.

- IaaS - Infrastructure as a Service

This model provides access to basic computing resources like virtual server, storage and networking over the internet. You are responsible for managing everything else including the operating system, applications & security, it offers maximum flexibility and control but requires you to be responsible

Example : Amazon Web Services (AWS) EC2 allows businesses to rent virtual servers to host websites, run databases or handle data processing without owning physical hardware

- PaaS - Platform as a Service

- The provider manages both the Infrastructure and a development platform that includes the operating system, runtime and tools
- You just focus on writing and deploying your code

Example - Google App Engine lets developers deploy code directly to Google's platform which automatically manages scaling, updates and server maintenance.

SaaS - Software as a Service

The provider delivers a complete, ready to use application via the internet, You don't handle any technical aspects, you just access and use the software through an app, The provider manages everything

Example : Microsoft 365 (Excel, Word, Outlook) is accessed online

26 Version Control System - A tool that tracks and manages changes to files overtime

- It maintains a complete history of all modifications
- Enables you to see what changed, who changed it and when

* Git is a widely used type of version control system developed in 2005 by Linus Torvalds, As a distributed system, it allows every person working on a project to have a complete copy of the entire project history on their own computer rather than solely on a central server

Why developers and Data analysts love Git?

- * It allows multiple Users to work on the same file at the same time (allows collaboration)
- * It can track who changed what, when and why
- * You can easily undo mistakes by rolling back to a previous working version

Commit - A saved snapshot of the project at a point in time
It includes a commit message explaining the change

Branch - An independent line of development off the main branch
- Allows you to build features / fix bugs without impacting the main code

27) Jupyter Notebook - An open source, interactive platform that lets you write and run code, see outputs and document your work all within a single file
- It is often used with Python

Why it is Popular among Data scientists and Analysts

- Context and Storytelling - Markdown supports lets users combine formatted text, code and visuals in one document, creating clear narratives that explain data insights alongside the analysis.
- Easy Reporting and collaborating - Notebooks can be exported as PDFs, webpages or presentations, making it simple to share detailed workflows and insights with colleagues / stakeholders
- Integrated Visualisation
- Code is divided into cells that can be run independently, which allows users to test and modify parts without rerunning the entire script

28) Python - A high level, open source programming language

- It emphasizes simplicity and readability with code that closely resembles plain English, making it easier to learn, write and maintain than ⁱⁿ any other languages
- It supports a wide range of applications, including web development, automation, data analysis, machine learning, AI etc.

- It is Beginner friendly (easy to understand)
- Free and Open Source - Accessible to individuals, students, start-ups, and enterprises without cost issues
- It is Versatile
- A large, active community ^{can} offers lots of resources and problem solving support

4 Essential Python Libraries for Data Work

1. NumPy (Numerical Computing)

- Core library for numeric data and scientific computing
- Introduces arrays for efficient handling of large numeric datasets

2. Pandas (Data Manipulation & Analysis)

- Handles structured data with its Dataframe format
- Makes loading, cleaning, filtering, transforming datasets easy
- Supports reading/writing CSV, Excel, SQL etc

3. Matplotlib (Data Visualization)

- Primary plotting library for creating charts & graphs
- Offers precise customization of visuals like line charts, bar charts

4. Scikit-learn (Machine Learning)

- Widely used library for building and evaluating predictive models
- Includes tools for classification, regression and data ~~prep~~ preprocessing

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Section D: Data Ethics, Governance & Careers

29 Data Privacy- The Right of Individuals to manage how their personal information is gathered, stored, utilized and shared by organizations and other entities

× It's important because it protects individuals from harm, improper handling of data that can result in identity theft, fraud, financial scams, and discrimination

× It also prevents surveillance and manipulation as personal information can be exploited

* The GDPR is a European regulation that sets mandatory standards for how organizations and companies must handle personal data in a way that respects privacy, its goal is to protect individuals and their personal information

× The GDPR applies to all 27 EU member countries along with 3 countries in the European Area (Norway, Iceland and Liechtenstein)

- The 27 EU member states include countries like Germany, France, Italy, Spain, Netherlands, Belgium, Poland, Sweden and others in Europe

30) Data Ethics - Refers to principles that ensure data is collected, stored and used honestly and respectfully

EXAMPLES of Unethical use of data or AI

1) Location Data Sold by free Apps - Free Apps may sell precise GPS data to data brokers without genuine informed consent, risking privacy and safety as this data can be misused for tracking or harm

2) Phone "Listening" - Apps may not record audio but gather extensive behavioural data to predict interests without clear user consent, turning private moments into commercial products

3) Job rejections based on social media - Employers or AI tools may screen personal social media, using posts or photos to unfairly reject candidates

- 4) Deep fakes / Generated Images and Videos of real people making it seem as if they said or did things they never actually did
- This can cause long lasting reputational harm since fake videos showing offensive/criminal behaviour can quickly go viral

3) Data Governance - collection of rules and processes that help keep an organization's data safe and accurate

Key Components of a Strong Data Governance framework

* People and how the organization is set up - data governance is all about human accountability, defining clear roles to manage data effectively

- Data Governance Council - Executive level stakeholders who set policies make huge decisions and solve high level data issues
- Data owners - Business leaders who hold ultimate accountability for their department's data and determine access's permission
- Data Stewards - Experts who handle day-to-day data management, and ensure that data remains accurate, high quality

* Rules, Standards & Processes

- Data Quality Standards - Criteria ensuring data is accurate, complete, consistent and timely
- Security & Privacy Policies - Guidelines for safe guarding data via classification and ensuring compliance with legal regulations like GDPR

* Data Architecture & Metadata Management

- Data Catalogue & Glossary - A searchable inventory of data assets paired with standardized definitions
- Data Lineage - A map that ~~tracks~~ tracks the journey of data, and show its origin, how it is modified and its final destination.

32) Bias in AI? - Bias in AI happens when an AI system gives unfair results meaning that the algorithm has learned to favor certain characteristics or a group of people over others

* If the data used to train the intelligence contains prejudices or reflects existing social problems the artificial intelligence will have those same biases

Example: In hiring and recruitment, the tools that screen job applications and generate job descriptions can be biased. They might prefer words that are typically associated with men, they might also penalize people who have gaps in their employment history, therefore affecting women who need the jobs and are qualified for them and people who might have taken breaks to start families (then took breaks) and now have gaps in their employment history

33) Business Intelligence is the way companies use technology to turn a lot of data into information that they can actually use

a) Microsoft Power BI - Lets people connect to a lot of data sources, build relationships between data and share reports with others in the company using the cloud, it is similar to Excel

b) Tableau - Tableau is very good at making data look nice. It is easy to use. Data analysts like to use it because it can handle a lot of data and let them make custom pictures and maps without having to write a lot of code

Microsoft Power BI and Tableau help companies make decisions by:

* Bringing all their data together in one place. (In most companies, different departments have their own data in separate places (different spreadsheets) for the sales team, marketing etc. So Power BI and Tableau bring all this data together in one place so everyone gains access to the information

* They also help by letting companies see what's happening in time - Tableau ^{and} Power BI let managers see what is happening now, example if sales are down in one area, the manager can see it and do something about it fast, instead of waiting for the report at the end of the month

- Power BI and Tableau make it easy to see what is going on
They can easily see which department is doing well / which products are not selling well, they make it easy to understand technical data
This saves a lot of time & money for the company

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- 5)

Section E : Personal Reflection

35) The concept I found interesting was the one about APIs and API keys - it is really interesting because they act as the digital handshake that allows 2 entirely different systems to communicate securely

- It is suprising to realize that behind every weather update on our phones or payment processed, there is a highly structured, gate kept conversation happening in seconds, The idea that a simple, unique string of characters (API Key) works as both a passport and security guard to grant permission for data exchange is a brilliant solution.

36 The concept I found difficult was JSON

At first, JSON felt like a foreign language - it was a bit confusing (the curly braces, nested brackets) with no logic, I struggled to understand how data was being structured.

* To understand this, I used W3schools for a few examples and practising, like writing out a fake product list in JSON format

- I then understood that JSON is just a way of storing and exchanging data like a neat labelled filing system

37) I see it affecting my everyday life by boosting my productivity
For example when I'm at work and there are certain concepts I don't understand, I can just ask it to explain them to me, making it easier to do my job or things like drafting emails

- So I see it simplifying my day to day task as an assistant

x It also helps me when it comes to decision-making processes, it can analyze concepts / data that I provide it with, then help me make decisions.

33) Key differences between a Data Analyst, Scientist, Engineer and AI Engineer

Role	What they do	Skills	Tools Used
Data analyst	Examines historical data to find trends then create reports for decision making	Data visualization, Statistics, business logic	SQL, Power BI, Tableau, Excel
Data scientist	Uses advanced maths and stats to build models that predict future outcomes	Machine learning, Heavy statistics, Predictive modelling	Python, Scikit-Learn, R, Jupyter
Data Engineer	Build pipelines and databases to collect and clean raw data	Data pipelines (ETL), cloud architecture, system scaling	SQL, Python, Spark, Kafka, AM, Airflow
AI Engineer	Takes AI tools models and deploys them into real world software and apps	Software Engineering, API integration, model optimization	Python, Docker, Kubernetes, LLM APIs